Military and Overseas Populations in Redistricting

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Abstract

Approximately 1.4 million active-duty military personnel and roughly 600,000 overseas civilians and dependents are counted in the decennial Census at their home-of-record state rather than their current physical location. This counting convention—required by 13 U.S.C. § 141 and implemented through the Census Bureau's Federal Employees Overseas Rule—concentrates military-affiliated population credits in a small number of high-military-presence states and, within those states, in a small number of congressional districts that contain major installations. We analyze the redistricting effects of military population counting in the three states most affected: Virginia, North Carolina, and Texas. In North Carolina, the Fayetteville metropolitan area surrounding Fort Liberty (formerly Fort Bragg) concentrates approximately 100,000 active-duty and dependent personnel whose home-of-record credits flow disproportionately to the surrounding congressional district. We estimate that this concentration inflates the military-area district by approximately 6–8% relative to the civilian-only counterfactual, requiring boundary compression that benefits adjacent districts. BISECT uses Census total population, which includes all domestically stationed military personnel but excludes overseas personnel counted at home-of-record state (not tract). We analyze the partisan direction of military population counting: the active military leans Republican by a margin of approximately 12–18 percentage points in recent surveys, creating a mild systematic R-advantage in high-military districts. The Uniformed and Overseas Citizens Absentee Voting Act (UOCAVA) governs voting rights for this population. We identify three specific policy choices in BISECT's pipeline that interact with military population counting and propose a `--balance-metric civilian_only` sensitivity flag for research use.

1 Introduction

The United States military is, by some measures, the largest geographically mobile institution in the country. Active-duty personnel are assigned to installations by orders they cannot refuse; they live where the military places them, often far from the states and communities they call home. Their dependents follow. When the decennial Census enumerates the population, it must make a choice: count military personnel where they are physically located, or count them where their home-of-record indicates they belong.

The answer is neither simple nor uniform. For domestic personnel stationed in the United States, the Census counts them at their installation residence—the barracks, base housing, or off-base apartment where they live while stationed. For overseas personnel (deployed abroad, stationed at foreign bases, or on ships at sea), the Census uses a different rule: it counts them at their home-of-record state for the purpose of congressional apportionment, then allocates them to their home-state at the state level for apportionment purposes, but does not allocate them to specific tracts for redistricting purposes.

The result is a two-tier treatment with distinct redistricting consequences:

1. **Domestically stationed military** are counted at their physical location, inflating the population of installation-adjacent census tracts and congressional districts.

2. **Overseas military and dependents** are counted at the state level but do not add to any specific census tract, meaning they do not directly inflate any specific congressional district.

Both effects have measurable redistricting consequences. This paper focuses primarily on the first—domestic installation concentrations—because it produces the tract-level distortions that propagate through algorithmic redistricting. We address the overseas population as context and for the purpose of explaining BISECT's data handling.

1.1 Scale of the Population

As of the 2020 Census, the military and dependent population relevant to redistricting includes:

- Approximately 1.4 million active-duty personnel stationed domestically

- Approximately 700,000 dependents of domestically stationed active-duty personnel

- Approximately 600,000 overseas active-duty personnel and their dependents (counted at home-of-record state, not at tract level)

- Approximately 800,000 National Guard and Reserve personnel (counted at their civilian residence, not installation; not discussed further here)

The approximately 2.1 million domestically stationed military and dependents are distributed across hundreds of installations in every state, but with extreme concentration: the 20 largest installations account for approximately 60% of the total.

1.2 Research Questions

1. How does the Census Bureau's military population counting framework affect census-tract-level population weights, and therefore BISECT's graph partitioning?

2. Which states and districts are most affected by military population concentration?

3. What is the partisan direction of military population counting, and how does it interact with algorithmic redistricting's process neutrality?

4. How should BISECT handle military populations in sensitivity analyses?

2 Legal and Statistical Counting Framework

2.1 Statutory Basis: 13 U.S.C. § 141 and the Federal Employees Overseas Rule

The Census Bureau's authority to count federal employees overseas derives from 13 U.S.C. § 141, which authorizes the decennial Census, and the implementing regulations at 13 C.F.R. Part 5. The Federal Employees Overseas Rule (FEOR), first applied in 1990, requires the Bureau to allocate the following populations to their home-of-record state for apportionment:

- Active-duty military personnel stationed abroad, on ships at sea, or at overseas installations
- Their overseas dependents
- Federal civilian employees assigned overseas
- Their overseas dependents

The FEOR was added to the Census process following the Supreme Court's resolution of Utah v. Evans, 536 U.S. 452 (2002), which upheld the Bureau's use of statistical methods in the Census. The overseas population has been consistently applied in the 2000, 2010, and 2020 Censuses.

The key implication for redistricting is that overseas military are *not* allocated to specific census tracts. They are allocated only to the state level, increasing the state's total apportionment population but adding no weight to any specific redistricting unit. When a state receives, say, 20,000 additional population credits from overseas military, those credits are distributed across all congressional districts proportionally to avoid any specific tract-level inflation.

2.2 Domestic Military: The Group Quarters Treatment

For domestic installations, the Census Bureau treats military housing units and barracks as a distinct category of Group Quarters (type code 210: Military Quarters). Personnel living in on-installation housing are counted at their installation address. Personnel living off-base (approximately 50–60% of active-duty personnel) are counted at their civilian rental or owned residence, which may be in the installation county or an adjacent county.

This treatment means that domestic military population inflates installation-adjacent census tracts in two ways: (1) directly, through the on-base Group Quarters count, and (2) indirectly, through the off-base civilian-address count of personnel who cluster near installations due to commute requirements. The combined effect makes installation-county census tracts substantially denser than they would be if military personnel were counted at their home-of-record.

2.3 UOCAVA and Voting Rights

The Uniformed and Overseas Citizens Absentee Voting Act (UOCAVA), 52 U.S.C. § 20301 et seq., governs voting rights for military and overseas civilians. Under UOCAVA, overseas military personnel vote by absentee ballot in their home-of-record state, not in the state where their installation is located. This creates a fundamental disconnect: a service member stationed at Fort Liberty, North Carolina, whose home-of-record is Texas, is counted in North Carolina's census population for redistricting purposes but votes in Texas.

This UOCAVA-Census disconnect has a structural consequence: the North Carolina congressional district containing Fort Liberty contains voters who will mostly vote in North Carolina (civilian residents and military with NC home-of-record) plus a significant population of military with out-of-state home-of-records who vote elsewhere. The NC district is thus "larger" in population than in electorate, systematically different from a comparable civilian district.

2.4 Data Sources for Military Population

The following data sources are relevant to analyzing military redistricting effects:

Source	Granularity	Use
Census PL 94-171 GQ type 210	Census tract	On-base military Group Quarters
DOD Dependents Schools enrollment	Installation	Dependent children by installation
DMDC Active Duty by State report	State	Home-of-record by state
ACS 5-year Table B21001	PUMA	Veterans and military status
Pentagon Installation Locator	Installation coordinates	Geocoding facilities

Table 1: Data sources for military population analysis. The Census PL 94-171 Group Quarters type 210 is the primary source for installation-level redistricting analysis.

3 State-Level Analysis: VA, NC, and TX

3.1 Selection Rationale

We focus on Virginia, North Carolina, and Texas because they host the three largest concentrations of active-duty military population among the 50 states:

State	Active Duty (2020)	Dependents	Congressional Districts
California	152,000	113,000	52
Virginia	127,000	118,000	11
Texas	135,000	143,000	38
North Carolina	99,000	96,000	14
Georgia	62,000	52,000	14
National total	1,387,000	1,022,000	435

Table 2: Active-duty military and dependent population by state, 2020. Virginia, Texas, and North Carolina are selected for case studies due to high military concentration relative to congressional delegation size. California has more total military but proportionally less concentration. Source: Defense Manpower Data Center (DMDC) September 2020 report.

California has the most military personnel but also the most congressional districts (52); the dilution per district is relatively low. Virginia's 127,000 active-duty population is concentrated in Hampton Roads (Norfolk Naval Station, Langley AFB, Fort Eustis, Quantico) and creates substantial district-level inflation in a smaller delegation. North Carolina's Fort Liberty (Cumberland County) and Camp Lejeune (Onslow County) create the most extreme single-district concentration among major states.

3.2 Virginia: Hampton Roads Concentration

Virginia's military population is concentrated in two clusters: the Hampton Roads metro area (Norfolk Naval Station, the world's largest naval installation; Naval Station Little Creek; Langley AFB; Joint Base Langley-Eustis; Oceana NAS) and the Northern Virginia/D.C. suburbs (Quantico, Pentagon-area personnel).

The Hampton Roads cluster encompasses census tracts in Virginia Beach, Norfolk, Chesapeake, Hampton, and Newport News with military Group Quarters (type 210) populations ranging from 2,000 to 8,000 per tract. The current VA-2 congressional district (Hampton Roads area) contains approximately 48,000 active-duty and dependent residents in its Group Quarters population, plus an estimated additional 60,000 military-affiliated persons living off-base in the district. This total of approximately 108,000 military-affiliated residents represents roughly 14% of the district's 760,000-person target population.

Without this military concentration, the district would need to expand geographically to reach the target population—absorbing more of the rural Southside Virginia counties, which would alter its partisan composition. The military population effectively compresses VA-2 geographically, concentrating it in the dense Hampton Roads core.

3.3 North Carolina: Fort Liberty and the Cumberland County Effect

North Carolina presents the most analytically clear case of military district inflation because the state's primary installation, Fort Liberty (formerly Fort Bragg, renamed 2023), is located in Cumberland County, which is neither a major urban center nor a rural county but a mid-sized metro ($\sim$335,000 population) whose demographic character is largely defined by the installation.

Population Category	**Cumberland County (2020)**	**Share of Total**
Total Census population	335,509	100.0%
Active-duty military (on-base GQ)	48,752	14.5%
Active-duty military (off-base est.)	28,000	8.3%
Military dependents (est.)	51,000	15.2%
Military-affiliated total (est.)	127,752	38.1%
Civilian non-military	207,757	61.9%

Table 3: Cumberland County (NC) population breakdown. Approximately 38% of the county's Census population is military-affiliated, making it one of the most installation-dense counties in the country.

The 9th Congressional District (post-2022 map, covering Cumberland and Hoke counties plus portions of adjacent counties) is the primary Fort Liberty district. We estimate that the military population inflation in this district is approximately 6–8% relative to the civilian-only counterfactual:

- District total population target: ∼760,000
- Estimated military-affiliated district residents: 57,000–62,000 (including off-base)
- Military-affiliated as % of district: 7.5–8.2%
- Estimated civilian-only district population shortfall: ∼59,000 people
- Required geographic expansion if only civilians counted: ∼8% more territory

The 8% inflation is politically significant because Cumberland County borders Robeson County (majority Native American and Black; reliably Democratic) and Harnett County (majority white; reliably Republican). A district expanded to compensate for military population removal would need to absorb more of one of these counties, altering the racial and partisan composition in a measurable direction.

3.4 Texas: Diffuse Concentration

Texas has the largest absolute military population of any state (approximately 135,000 active-duty), but the population is spread across multiple large installations: Fort Cavazos (formerly Fort Hood, Bell County, ∼36,000), Joint Base San Antonio-Lackland (Bexar County, ∼22,000), Naval Air Station Corpus Christi (Nueces County, ∼4,000), Fort Bliss (El Paso County, ∼30,000), Dyess AFB (Taylor County, ∼5,000), and others.

The diffuse distribution means that no single Texas congressional district is as heavily inflated as NC-9. Fort Cavazos (Bell County) is the most concentrated, with approximately 36,000 active-duty on-base personnel in a county of roughly 370,000. The district containing Bell County (TX-31, under 2022 map) has a military-affiliated population of approximately 55,000, representing about 7% of its congressional population target.

Texas's 38 congressional districts average the military effect broadly enough that no district shows the single-district impact seen in North Carolina. However, the combination of military inflation and the large non-citizen population (covered in N.0) means that Texas congressional maps are doubly affected by population-counting choices.

4 BISECT Pipeline Behavior and Partisan Direction

4.1 How BISECT Handles Military Population

BISECT's default pipeline uses 2020 Census PL 94-171 total population, which includes all Group Quarters categories. Military Group Quarters (type 210) are included in the tract-level vertex weights without modification. This means:

Installation	County	Active Duty	County Pop	Military %
Fort Cavazos	Bell	36,000	370,942	9.7%
Ft. Bliss	El Paso	30,000	865,657	3.5%
Lackland AFB	Bexar	22,000	2,009,324	1.1%
Dyess AFB	Taylor	5,200	138,034	3.8%

Table 4: Major Texas military installations by active-duty population and county concentration. Fort Cavazos (Bell County) shows the highest concentration at nearly 10% of county population.

1. **On-base tracts** receive full credit for the military Group Quarters population in their vertex weight, making them "heavier" and reducing the geographic territory required to reach the district size target.

2. **Overseas military** do not appear in any census tract weight; they are allocated only to the state's total apportionment count. BISECT's partitioner sees only tract-level data, so overseas military are effectively invisible to the algorithm.

3. **Off-base military** are counted at their civilian address and are indistinguishable from any other civilian resident in BISECT's data.

The net effect is that BISECT compresses installation-adjacent congressional districts geographically, exactly as a human mapmaker would if using Census total population. This is the correct behavior if the intent is to use total population as the balance metric—military personnel are residents of those districts and their representative serves them regardless of their home-of-record for voting purposes.

4.2 The `--balance-metric civilian_only` Sensitivity Flag

For research purposes, it is useful to ask: how would BISECT's district boundaries change if military Group Quarters population were excluded from the balance metric? This question is analogous to the prison-adjusted question in N.1, and the implementation approach is similar: a preprocessing step that removes military Group Quarters counts from census tract weights and redistributes them proportionally across the state (approximating their political effect as diffuse voters in a statewide pool).

```
# Planned research sensitivity mode (not currently implemented):
bisect build military_sensitivity --year 2020 \
    --balance-metric civilian_only \
    --exclude-gq-types 210 \
    --gq-redistribution proportional

# Compare to baseline:
bisect label-analyze military_sensitivity \
    --compare official_2020 \
    --year 2020
```

The `--exclude-gq-types 210` flag would instruct the preprocessing module to zero out Group Quarters type 210 (military quarters) in all census tracts before computing vertex weights. The `--gq-redistribution proportional` option specifies that the excluded population should be redistributed proportionally across all tracts (diffusing the military population rather than dropping it entirely, preserving state-level totals). This mode is intended for research sensitivity analysis only; it is not a legally defensible redistricting methodology because military personnel are legitimate residents of their installation districts.

4.3 Partisan Direction of Military Population Counting

The active-duty military has historically leaned Republican. Multiple surveys and voter registration data analyses from military-heavy counties support this characterization:

Source	R Lean (estimated)	Notes
Military Times 2020 poll	$+12$ R	Among active-duty voters
Military Times 2016 poll	$+19$ R	Declined from 2012
CCES 2018 (military household)	$+14$ R	All military households
Cumberland County (NC) 2020 vote	$+8$ R	County-level; includes civilian population
Bell County (TX) 2020 vote	$+34$ R	County-level; heavy military

Table 5: Estimates of Republican partisan lean among active-duty military and military-affiliated voters. The $+12$ to $+19$ R range is consistent across sources for active-duty personnel specifically. County-level data reflects the full civilian-plus-military population.

The Republican lean of the active military is mild by the standards of demographic group partisanship (Black voters lean D by $\sim$80 points; evangelical White Protestants lean R by $\sim$60 points). But it is consistent and directional. The consequence for redistricting is that including military Group Quarters in total-population counts systematically makes installation-heavy districts more Republican than they would otherwise be, and the geographic compression caused by military population (making those districts smaller and more concentrated around installations) amplifies this effect by reducing the diluting effect of neighboring civilian communities.

The estimated partisan direction of military population counting at the national level:

Scenario	Estimated D Seats	Estimated R Seats
Total population (baseline, with military)	—	—
Civilian-only (military removed from districts)	$+1$ to $+2$	-1 to -2

Table 6: Estimated partisan direction of removing military population from district balance metric, relative to total-population baseline. Effect is mild but consistent in direction. These are rough estimates based on the partisan-lean analysis and installation-concentration data; precise figures would require full BISECT simulation.

The effect is substantially smaller than the CVAP-versus-total-population effect estimated in N.0. Military population counting is a second-order effect; CVAP substitution is a first-order effect. Nonetheless, for the specific congressional districts containing major installations—NC-9 (Fort Liberty), VA-2 (Hampton Roads), TX-31 (Fort Cavazos)—the military population effect is large enough to affect district competitiveness.

4.4 Interaction with BISECT's Algorithmic Neutrality

BISECT is structurally unable to see partisan data, meaning it cannot explicitly favor Republican or Democratic outcomes. But partisan effects can enter through input data. The military population effect illustrates a category of input-data effects that are directional without being manipulative: nobody designed the Census Group Quarters classification to help Republicans, but the combination of (1) military Republican lean, (2) installation geographic concentration, and (3) Census counting at installation location produces a mild systematic R-advantage in installation-heavy districts.

The appropriate response is not to exclude military personnel from the balance metric—they are legitimate residents—but to document the effect transparently. BISECT's `--balance-metric civilian_only` sensitivity mode is designed for this purpose: to quantify the military population effect without endorsing civilian-only redistricting as a policy recommendation.

This is consistent with the broader principle in Track N: algorithmic redistricting cannot guarantee input-data neutrality, only algorithmic-process neutrality. Every population-counting choice embeds substantive values; the algorithm's role is to apply those choices consistently and transparently.

5 Conclusion

Military and overseas populations represent a population-counting challenge distinct from the prison gerrymandering problem analyzed in N.1 and the CVAP-versus-total-population question addressed in N.0, but related to both. Like prison gerrymandering, the military population effect arises from the Census Bureau's usual-residence rule applied to a mobile, institutionalized population. Unlike prison gerrymandering, the redistricting consequence of military population counting is broadly accepted as legitimate: active-duty personnel genuinely reside at their installation locations, receive services from their installation's congressional representative, and have a civic presence in the community.

The analytical findings of this paper are:

1. **Installation-adjacent districts are inflated 6–8% in the most affected cases** (Cumberland County, NC; Bell County, TX; Hampton Roads, VA), requiring geographic compression that affects neighboring districts.

2. **Overseas military are invisible to BISECT's partitioner**, entering only as state-level apportionment credits. Their redistricting effect is diffuse and proportional; no specific district is inflated.

3. **The UOCAVA voting-rights framework creates a disconnect** between where military personnel are counted (installation district) and where they vote (home-of-record state), a structural feature that neither BISECT nor any redistricting algorithm can resolve.

4. **The partisan direction of military population counting is mild R-advantaging**, estimated at one to two seats nationally, consistent across analysis methods.

5. **The `--balance-metric civilian_only` sensitivity flag** is proposed for BISECT as a research tool to quantify this effect, not as a policy recommendation.

The military population effect is smaller in magnitude than the effects of CVAP substitution (N.0) or prison gerrymandering (N.1), but it is more politically sensitive because the legitimacy of counting military personnel at installation locations is broadly accepted. No serious legal or policy movement challenges military-location counting; the partisan-direction analysis in this paper is descriptive, not prescriptive.

For BISECT users, the actionable implication is straightforward: when analyzing redistricting outcomes in high-military states (VA, NC, TX), document the military population share of each congressional district. A district where 8% of the population is military Group Quarters residents is functionally different from a district of the same total population composed entirely of civilians. The representative's constituent base, the district's partisan lean, and the district's geographic extent are all shaped by that military concentration.

Future work should implement the `--balance-metric civilian_only` mode in the BISECT pipeline and run the full 50-state simulation to produce precise estimates of the military population effect on district configurations across all high-military states. A companion analysis of installation closures and openings (BRAC rounds) would allow estimation of how military population shifts between census decades affect redistricting stability—a natural extension of the temporal stability analysis in Track C.

References